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Assignment number and name: Task 2 – Custom IAM Policy, EC2 Role & MFA

Submission date: 19 September 2026

AWS Region Used: us-east-1(N. Virginia)

Task – Custom IAM Policy, EC2 Role & MFA Configuration

Objective

Write a custom IAM policy, create an EC2 role using the policy, and enable MFA on an IAM user.

What You Need To Do

Go to IAM → Policies → Create Policy → JSON editor.
Write a custom policy that:
- Allows s3:ListAllMyBuckets
- Allows s3:GetObject
- Explicitly denies s3:DeleteObject

Name the policy:
S3-ReadOnly-NoDeletion

Go to IAM → Roles → Create Role.
Trusted entity:
AWS Service → EC2
Attach policy:
S3-ReadOnly-NoDeletion
Name the role:
EC2-S3-Reader
Create role.

Go to IAM → Users → dev-user → Security Credentials.
Assign MFA device.
Select Virtual MFA device.
Scan QR code with Google Authenticator.
Complete setup.
Log out.

Log in as dev-user.
Confirm MFA prompt appears before console access.

Log back in as root/admin.
Go to IAM → Roles → EC2-S3-Reader → Trust Relationships tab.
Review the trust policy JSON.
Note:
ec2.amazonaws.com as the trusted service.

Go to IAM → Policies → S3-ReadOnly-NoDeletion → Policy summary.
Verify:
- Allowed actions
- Explicitly denied action

Free Tier Notice:
IAM roles, policies, and MFA are completely free.

Deliverables

Screenshots of:
- Custom policy JSON
- Policy summary page
- IAM role summary
- Trust policy JSON
- dev-user MFA device confirmation
- MFA prompt during login

Steps Performed

Step 1 – Custom IAM Policy

Action

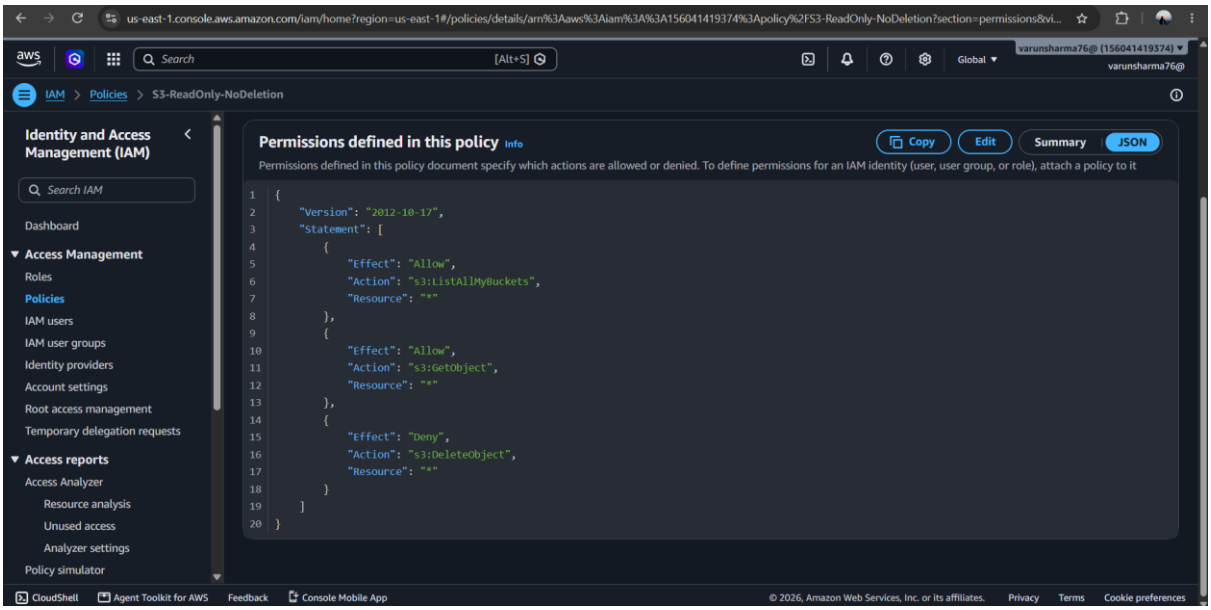
Navigated to IAM → Policies → Create policy → JSON editor and created the required custom S3 policy.
Policy name: S3-ReadOnly-NoDeletion
Result: The custom policy was created successfully.

Custom Policy JSON

```
{
"Version": "2012-10-17",
"Statement": [
{
```

```
"Effect": "Allow",
"Action": "s3:ListAllMyBuckets",
"Resource": "*"
},
{
"Effect": "Allow",
"Action": "s3:GetObject",
"Resource": "*"
},
{
"Effect": "Deny",
"Action": "s3:DeleteObject",
"Resource": "*"
}
]
}
```

Confirmation



Step 2 – Policy summary verification

Action

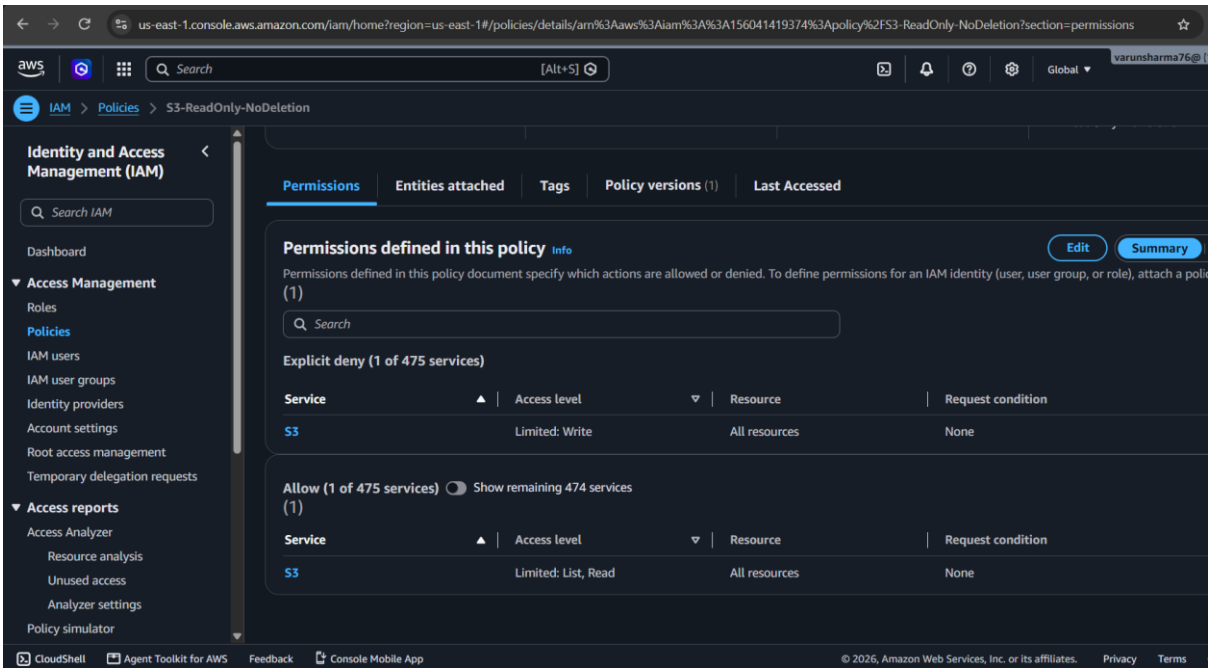
Opened S3-ReadOnly-NoDeletion and reviewed its Policy Summary.

Allowed actions: s3:ListAllMyBuckets and s3:GetObject.

Explicitly denied action: s3:DeleteObject.

Result: The policy summary showed S3 List/Read access and an explicit deny for S3 write/delete access.

Confirmation



Step 3 – Created EC2 IAM Role

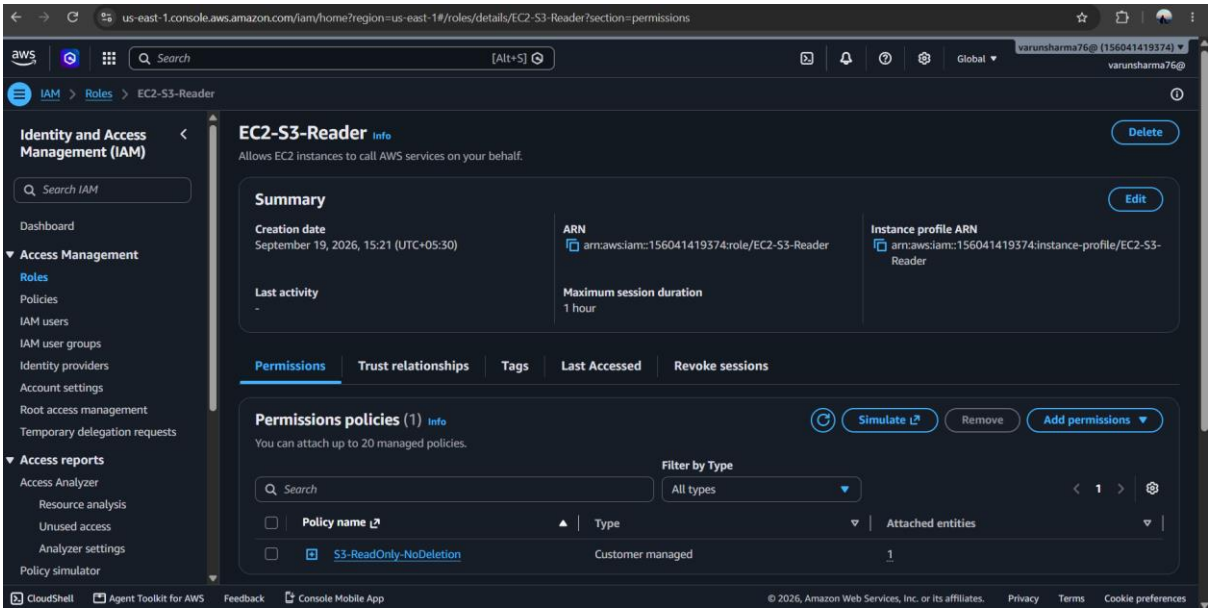
Action

Navigated to IAM → Roles → Create role, selected AWS Service → EC2 as the trusted entity, attached S3-ReadOnly-NoDeletion, and created the role.

Role name: EC2-S3-Reader

Result: The EC2 role was created and the custom policy was attached.

Confirmation



Step 4 – Trust Relationship Verification

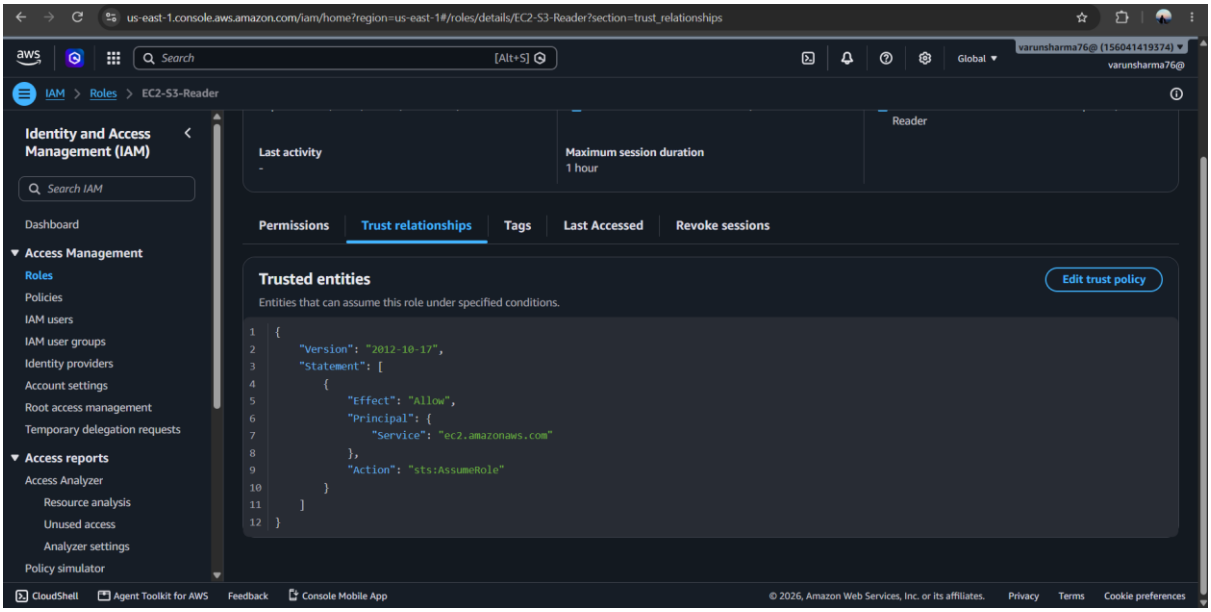
Action

Opened EC2-S3-Reader → Trust relationships and reviewed the trust policy JSON.

Result: The EC2 service was configured as the trusted service and was allowed to assume the role.

{
"Version": "2012-10-17",
"Statement": [
{
"Effect": "Allow",
"Principal": {
"Service": "ec2.amazonaws.com"
},
"Action": "sts:AssumeRole"
}
]
}

Confirmation



Step 5 – MFA Device Conformation

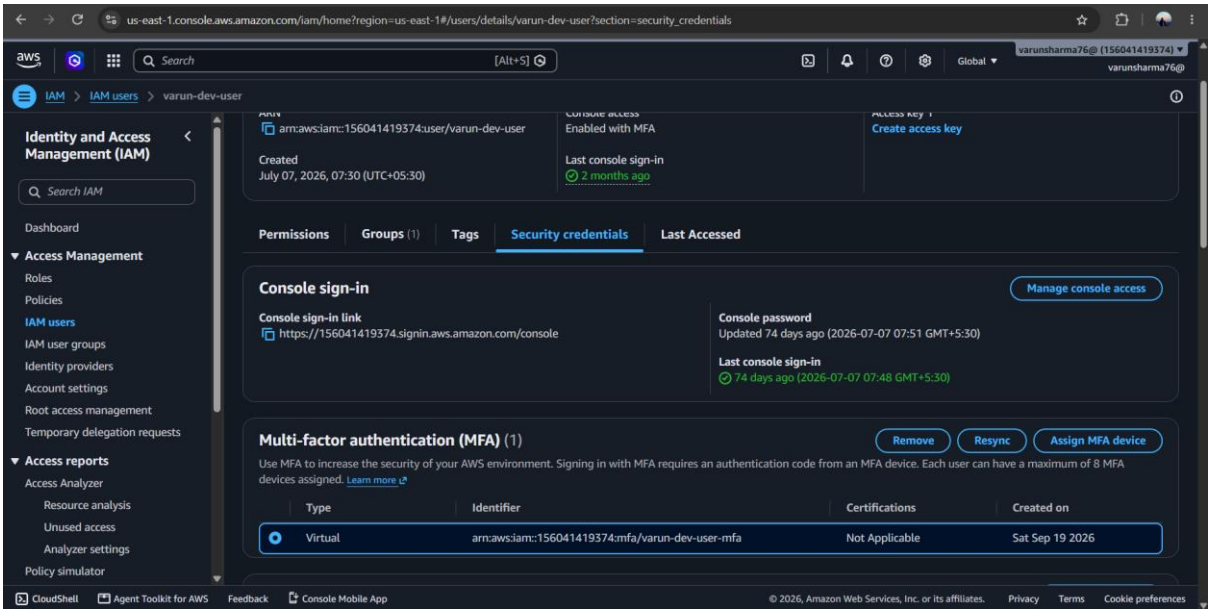
Action

Verified the existing IAM user used for the task, varun-dev-user, in IAM → Users → Security credentials. A Virtual MFA device was assigned and the console access status showed that MFA was enabled.

Note: The account already contained an IAM user named varun-dev-user from earlier practice tasks; this existing user was used for the Task 2 MFA verification rather than creating a duplicate user.

Result: Console access was shown as enabled with MFA and a Virtual MFA device was present.

Confirmation



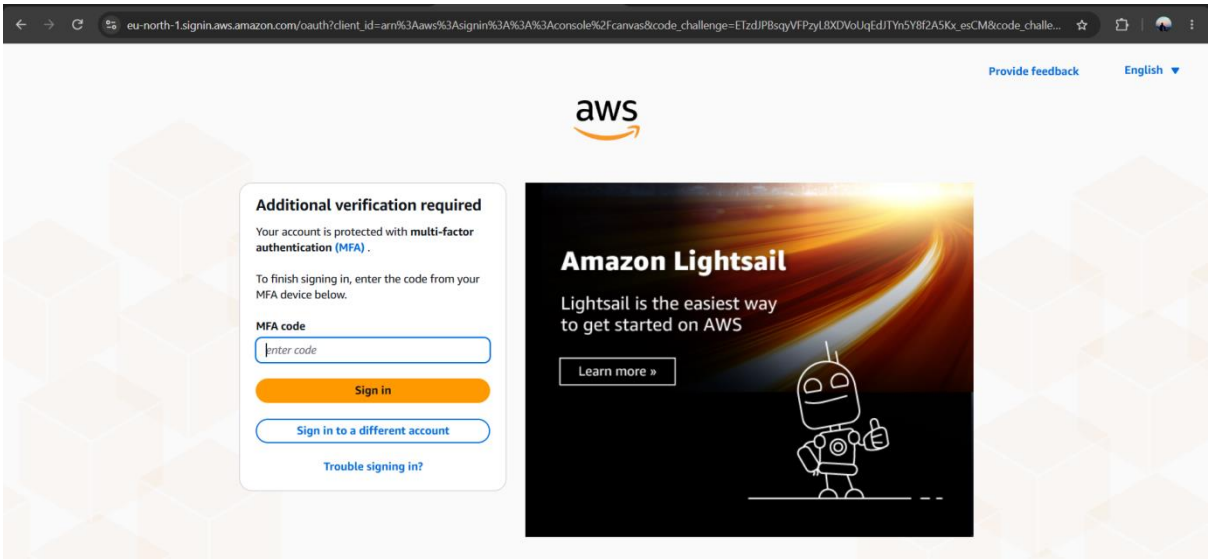
Step 6 – MFA Login Verification

Action

Signed out and logged in as varun-dev-user using the IAM user credentials. AWS requested an MFA verification code before console access.

Result: The MFA prompt appeared and the login was completed using the Google Authenticator code.

Confirmation



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7. AWS Resource Information

Resource	Name	ARN / Resource ID
Custom IAM Policy	S3-ReadOnly-NoDeletion	[PASTE ACTUAL POLICY ARN HERE]
EC2 IAM Role	EC2-S3-Reader	arn:aws:iam::156041419374:role/EC2-S3-Reader
IAM User	varun-dev-user	arn:aws:iam::156041419374:user/varun-dev-user

8. Final Result

The custom IAM policy S3-ReadOnly-NoDeletion was successfully created and verified. The EC2-S3-Reader role was created with the custom policy attached, and its trust relationship was verified for ec2.amazonaws.com. MFA was successfully configured for varun-dev-user and the MFA prompt was verified during IAM user login.